Logic and Set Theory

Flag-style derivations

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Introduction of bi-implication

How do we establish a bi-implication in a reasoning?

$(P \Rightarrow Q) \wedge (Q \Rightarrow P) \overset{val}{\models} P \Leftrightarrow Q$

⇔-introduction:

$$
\begin{array}{|ll|}
\hline
 & \vdots \\
(k) & P \Rightarrow Q \\
 & \vdots \\
(\ell) & Q \Rightarrow P \\
 & \{\Leftrightarrow\text{-intro on } (k) \text{ and } (\ell)\text{:}\} \\
(m) & P \Leftrightarrow Q \\
\hline
\end{array}
$$

$(\uparrow)$ $\qquad\qquad$ $(k, \ell < m)$

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Elimination of bi-implication

How can we use a bi-implication in a reasoning?

$$P \Leftrightarrow Q \models^{val} \begin{cases} P \Rightarrow Q \\ Q \Rightarrow P \end{cases}$$

$\Leftrightarrow$-elimination:

	$\Vert\Vert$
(k)	$P \Leftrightarrow Q$
	$\Vert\Vert$
	$\{\Leftrightarrow$-elim on $(k)\colon\}$
(ℓ)	$P \Rightarrow Q$

$(\downarrow)$ $(k < \ell)$

	$\Vert\Vert$
(k)	$P \Leftrightarrow Q$
	$\Vert\Vert$
	$\{\Leftrightarrow$-elim on $(k)\colon\}$
(ℓ)	$Q \Rightarrow P$

$(\downarrow)$ $(k < \ell)$

Example

Show with a *flag-style derivation* (according to the methods of part II of the book) that the formula

$$(P \Leftrightarrow Q) \Rightarrow ((P \wedge Q) \vee (\neg P \wedge \neg Q))$$

is a tautology.

[See the video 21 Disjunction and bi-implication (example).]

Example

Show with a *flag-style derivation* (according to the methods of part II of the book) that the formula

$$((P \vee Q) \wedge \neg(Q \Leftrightarrow R) \wedge R) \Rightarrow P$$

is a tautology.

An example of a mathematical proof

Theorem
For all $x \in \mathbb{Z}$, if x^2 is odd, then so is x.

Proof:
Let $x \in \mathbb{Z}$, and suppose x^2 is odd;
we need to prove that x is odd too.

To this end, suppose that x is even;
we derive a contradiction.

If x is even, then $x = 2y$ for some $y \in \mathbb{Z}$.
Then $x^2 = (2y)^2 = 4y^2 = 2(2y^2)$ and $2y^2 \in \mathbb{Z}$.

So x^2 is even, and we have our contradiction. □

Proof by contradiction

Assume:

(k) $\boxed{\neg P}$

$\vdots$

$(\ell - 1)$ `False`

{ RAA* on (k) and $(\ell - 1)$: }

(ℓ) P

* RAA: Reductio Ad Absurdum.

Book does not use this name, but introduces rules for $\neg\neg$ instead (see 14.4, 14.5). Both ways are fine.

Example

Show with a *flag-style derivation* (according to the methods of part II of the book) that the formula

$$(\neg P \Rightarrow (P \wedge Q)) \Rightarrow P$$

is a tautology.

[See the video 19 Proof by contradiction (example).]

Constructing a derivation

forward reasoning: draw conclusions from hypotheses or facts; and

backward reasoning: simplify one of the remaining goals (on basis of main symbol)

Main strategy:

Stick to backward reasoning until no remaining goals can be simplified further; then consider forward reasoning.

Main strategy works most of the time, and should always be tried first.

Fall-back strategies:

1. invent and prove new goal to be able to use hypotheses; or
2. apply a proof by contradiction.

Use fall-back strategies with caution; **only** if main strategy does not work!

Case distinction

How can we prove the inequality $|x-1| \geq \frac{1}{2}x - \frac{1}{2}$ for all $x \in \mathbb{Z}$?

Consider an arbitrary $x \in \mathbb{Z}$. Note that

$$|x-1| = \begin{cases} x-1 & \text{if } x \geq 1 \quad \text{(case 1)} \\ -x+1 & \text{if } x \leq 1 \quad \text{(case 2)} \end{cases}.$$

We therefore distinguish two cases:

1. If $x \geq 1$, then $\frac{1}{2}x \geq \frac{1}{2}$, so $|x-1| = x-1 \geq \frac{1}{2}x - \frac{1}{2}$.
   (To see that the last inequality holds, add $\frac{1}{2}x - 1$ on both sides in $\frac{1}{2}x \geq \frac{1}{2}$.)
2. If $x \leq 1$, then $-\frac{3}{2}x \geq -\frac{3}{2}$, so $|x-1| = -x+1 \geq \frac{1}{2}x - \frac{1}{2}$.
   (To see that the last inequality holds, add $\frac{1}{2}x + 1$ on both sides in $-\frac{3}{2}x \geq -\frac{3}{2}$.)

In both cases the inequality holds, and (at least) one of the two cases applies for every $x \in \mathbb{Z}$.
So the inequality holds for every $x \in \mathbb{Z}$.

Case distinction (rule)

Proof by case distinction:

$$
\begin{array}{ll}
 & \| \| \\
(k) & P \vee Q \\
 & \| \| \\
(\ell) & P \Rightarrow R \\
 & \| \| \\
(m) & Q \Rightarrow R \\
 & \| \| \\
 & \{ \text{Case distinction on } (k), (\ell) \text{ and } (m): \} \\
(n) & R
\end{array}
$$

Introduction of double negation

How can we establish $\neg\neg$ in a reasoning?

$P \overset{val}{\models} \neg\neg P$

$\neg\neg$-introduction:

$$
\begin{array}{ll}
 & \vdots \\
(\ell - 1) & P \\
 & \{ \neg\neg\text{-intro on } (\ell - 1): \} \\
(\ell) & \neg\neg P
\end{array}
$$

$(\uparrow)$

Elimination of double negation

How can we use $\neg\neg$ in a reasoning?

$\neg\neg P \models^{val} P$

$\neg\neg$-elimination:

$$
\begin{array}{|ll|}
\hline
 & \| \| \\
(k) & \neg\neg P \\
 & \| \| \\
 & \{\ \neg\neg\text{-elim on } (k)\text{: }\} \\
(\ell) & P \\
\hline
\end{array}
$$

$(\downarrow)$

Example

Show with a *flag-style derivation* (according to the methods of part II of the book) that the formula

$$(P \Rightarrow Q) \Rightarrow (\neg P \vee (\neg Q \Rightarrow R))$$

is a tautology.